AutoCare

Online Car Service Booking System

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Project Guide
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Introduction

AutoCare is a comprehensive web-based car service booking platform that revolutionizes the traditional car servicing experience. Built with modern technologies, it enables users to seamlessly book various car services online, track their bookings, and manage their service history through an intuitive interface.

Domain

Web Application

Full-Stack Development

Technology

Spring Boot

MongoDB • Thymeleaf

Users

Car Owners

Service Administrators

Problem Statement

What problem are we solving?

- Traditional car servicing requires physical visits and phone calls for booking
- Lack of transparency in service tracking and pricing
- Time-consuming manual processes for both customers and service centers
- No centralized system to manage booking history and service records

Why is it important?

- Saves valuable time for car owners by eliminating unnecessary visits
- Improves service center efficiency through automated booking management
- Provides better customer experience with 24/7 booking availability

Project Objectives

1. Develop Online Booking

Create an intuitive web platform for customers to book car services online anytime, anywhere

2. Real-time Management

Implement real-time booking tracking and management system for both users and administrators

3. Secure Data Storage

Utilize MongoDB for reliable and scalable storage of user data, bookings, and service history

4. User-Friendly Interface

Design responsive and accessible UI using Thymeleaf templates for optimal user experience

Existing System

Current Approach

- Manual booking through phone calls
- Physical visit to service centers
- Paper-based record keeping
- Limited service hours availability

These limitations create a need for a modern, automated solution that can streamline the entire car service booking process.

Major Limitations

⚠ **Time-consuming booking process**

⚠ **No booking history tracking**

⚠ **Prone to human errors**

⚠ **Limited scalability**

⚠ **Poor customer experience**

Proposed System

AutoCare Solution

A comprehensive web-based platform that digitizes the entire car service booking ecosystem, providing seamless experience for both customers and service providers.

Key Improvements

24/7 Online Booking

Book services anytime from anywhere

Real-time Tracking

Monitor booking status instantly

Automated Management

Reduce manual work and errors

Centralized Database

Secure storage of all records

Technologies Used

Backend

- Language: Java 17
- Framework: Spring Boot 3.2.0
- Build Tool: Apache Maven 3.9.x

Database

- Database: MongoDB 7.0
- Type: NoSQL Document Store
- Connection: mongodb://localhost:27017

Frontend

- Template Engine: Thymeleaf
- UI: HTML5, CSS3, JavaScript
- Design: Responsive Dark Theme

Development Tools

- IDE: VS Code / IntelliJ IDEA
- Server: Embedded Tomcat
- Port: 8080

Architecture: Model-View-Controller (MVC) Pattern

System Architecture & Database Design

MVC Architecture

View Layer (Thymeleaf Templates)

Controller Layer (Spring MVC)

Service Layer (Business Logic)

Repository Layer (Data Access)

MongoDB Database

Database Entities

Customer

customer_id, name, email, phone

Booking

booking_id, customer_id, car_id, booking_date, notes

Car & Service

car_id, brand, model | service_id, name

Working Methodology

User Booking Flow

1 User Access → 2 Select Service → 3 Enter Details → 4 Submit Booking

System Processing

Validate Input → Save to MongoDB → Update Display → Confirmation to User

Admin Management Flow

- View all bookings in real-time dashboard
- Update booking status (Pending → In Progress → Completed)
- Manage customer information and service records
- Generate reports and analytics

Key Features

User Management

Registration, login, and profile management with secure authentication

Service Booking

Easy selection of services with car details input and instant booking

Real-time Tracking

Live updates on booking status and service progress visualization

Admin Dashboard

Comprehensive control panel for managing bookings and customers

Responsive Design

Mobile-friendly interface accessible on all devices

Data Persistence

Secure MongoDB storage with automatic data backup

Implementation Overview

User Interface Highlights

Home Page

Dark-themed landing page with service booking form and real-time booking display

User Dashboard

Personalized dashboard showing booking history, upcoming services, and account details

Admin Panel

Comprehensive admin interface for managing all bookings, users, and generating reports

Authentication System

Secure login and registration pages with form validation and error handling

Results & Achievements

What We Achieved

100%	24/7	Real-time	MVC
Functional System	Availability	Processing	Architecture

Key Outputs

- ✓ Fully functional web application deployed on localhost:8080
- ✓ Integrated MongoDB database with automated data persistence
- ✓ Responsive UI compatible with desktop and mobile devices
- ✓ Complete CRUD operations for booking management
- ✓ Secure user authentication and session management
- ✓ Admin dashboard with comprehensive controls

Current Limitations

Technical Constraints

1. Payment Gateway Integration

No online payment system currently implemented

2. Email Notifications

Automated email alerts not yet configured

3. Advanced Analytics

Limited reporting and business intelligence features

4. Multi-language Support

Currently available only in English

5. Service Provider Network

Limited to single service center management

6. Cloud Deployment

Runs on localhost, not yet cloud-hosted

Note: These limitations represent opportunities for future enhancement and do not affect the core functionality of the current system.

Future Scope & Enhancements

Planned Improvements

Payment Integration

Implement Razorpay/Stripe for secure online payments and transaction history

AI-Powered Recommendations

Smart service suggestions based on car model, usage, and service history

Mobile Application

Native Android and iOS apps with push notifications and GPS tracking

Multi-Center Network

Expand to support multiple service centers with location-based search

Advanced Analytics Dashboard

Data visualization, revenue reports, and predictive maintenance alerts

Cloud Deployment (AWS/Azure)

Scalable cloud hosting with CI/CD pipeline and auto-scaling

References

Technologies & Resources

- Spring Framework Documentation
https://spring.io/projects/spring-boot

- MongoDB Official Documentation
https://www.mongodb.com/docs/

- Thymeleaf Template Engine Guide
https://www.thymeleaf.org/documentation.html

- Apache Maven Build Tool
https://maven.apache.org/guides/

- GitHub Repository
https://github.com/yashk2004/Online-car-service.git

Thank You

AutoCare - Online Car Service Booking System

Any Questions?